

An Introductory Overview of Natural Language Processing: Evaluating Local SLMs, Multilingual Detection, and PII Extraction

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Abstract— Natural Language Processing (NLP) converts unstructured human text into structured numerical representations suitable for computational intelligence. This report documents the practical evaluation of fundamental NLP techniques using client-side, browser-based AI playgrounds across three core tasks: generative abstractive summarization using local Small Language Models (SLMs), multilingual text classification, and automated Personally Identifiable Information (PII) extraction. Experimental results confirm that local SLMs like Microsoft Phi 3.5 Mini can synthesize complex technical documents directly in the browser without cloud dependencies. Multilingual detection demonstrated high classification confidence (98%) on single-script inputs, but exposed significant confidence degradation (19%) on multilingual inputs, reinforcing the requirement for sentence-level pre-processing. Finally, automated PII extraction demonstrated effective privacy sanitization while revealing Named Entity Recognition (NER) false positives on corporate brand names. These insights highlight key trade-offs between sparse lexical frequency (TF-IDF) and dense semantic embeddings, establishing essential architectural strategies for upcoming course milestones and capstone system development.

Keywords— *Natural Language Processing (NLP), Small Language Models (SLMs), Abstractive Summarization, Multilingual Classification, Named Entity Recognition (NER), Semantic Embeddings.*

I. INTRODUCTION

Natural Language Processing (NLP) converts unstructured text into numerical representations that computational systems can interpret [1]. Raw text first undergoes tokenization and pre-processing, including text normalization, stop-word removal, n-gram extraction, stemming, lemmatization, and parts-of-speech (POS) tagging, to clean inputs and preserve context [2].

Beyond basic text cleaning, statistical models like Term Frequency - Inverse Document Frequency (TF-IDF) evaluate keyword importance across document collections [3], while graph-based algorithms like TextRank extract key text segments [4]. Modern NLP advances beyond strict word frequency by projecting tokens into dense vector embeddings, using Cosine Similarity to measure semantic proximity between concepts.

To bridge theoretical concepts with practical execution, this report documents the Unit 5 text analytics laboratory activities using client-side, browser-based AI playgrounds. The experiments evaluate three core NLP tasks: generative abstractive summarization using local Small Language Models (SLMs), multilingual text classification, and

automated PII entity extraction for privacy compliance [5], [6].

II. EXPERIENTIAL & PRACTICAL INSIGHTS

In this section, we discuss the practical results and findings from the Unit 5 text analytics laboratory activities. The experiments were conducted using client-side, browser-based AI playgrounds across three specific NLP tasks.

A. Generative Abstractive Summarization

The first task evaluated local Small Language Model (SLM) performance using Microsoft Phi 3.5 Mini in a browser-based chat playground. The model was set up with the system instruction, "You are an AI assistant that analyzes and summarizes text," and was tasked with processing a historical technical review of the Commodore 64 computer.

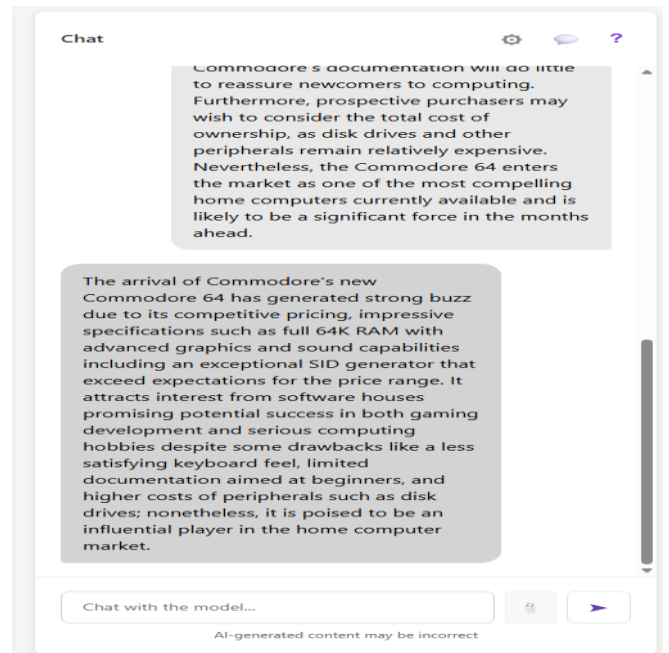


Fig. 1. Generative Abstractive Summary by Microsoft Phi 3.5 Mini.

As shown in Fig. 1, the model performed abstractive summarization by rewriting key information into a single cohesive summary paragraph. It successfully combined positive features like competitive pricing, 64K RAM, and the SID sound generator with drawbacks such as a stiff keyboard feel and expensive disk drives. By synthesizing these details instead of copying verbatim sentences, the model demonstrated that local small language models can perform

accurate text summarization directly in the browser without relying on external cloud servers.

B. Multilingual Text Classification

The second task evaluated automated language detection using the Analyzer tool in the browser playground. Two distinct language samples were tested to evaluate script recognition and text structure identification.

A sample input containing a multi-sentence Japanese paragraph was processed to test script recognition.

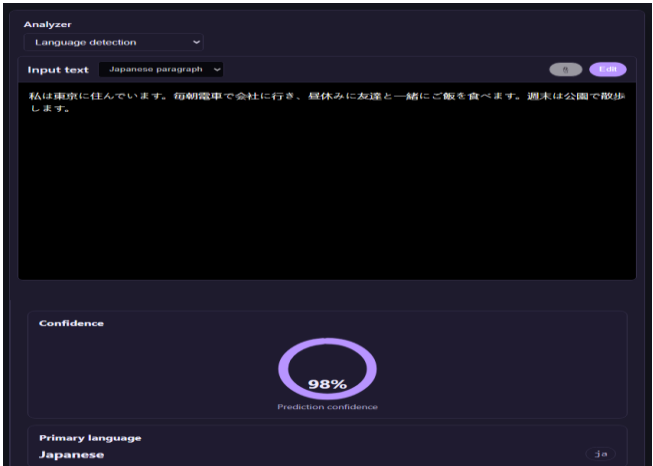


Fig. 2. Japanese language detection output

As shown in Fig. 2, the system correctly classified the text as Japanese with a high prediction confidence score of 98%. The model successfully recognized script-specific character sets and n-gram patterns to distinguish the non-Latin script from other languages. This exercise demonstrated that automatic language detection serves as an essential pre-processing step in text analytics pipelines, ensuring unstructured documents are correctly identified before being routed to language-specific translation or processing tools.

A second test analyzed text extracted from a vintage computer casing label (Schneider CPC 464). The input contained serial numbers, technical electrical values, and German terms such as "Hergestellt in Korea" and "Bundesrepublik Deutschland".

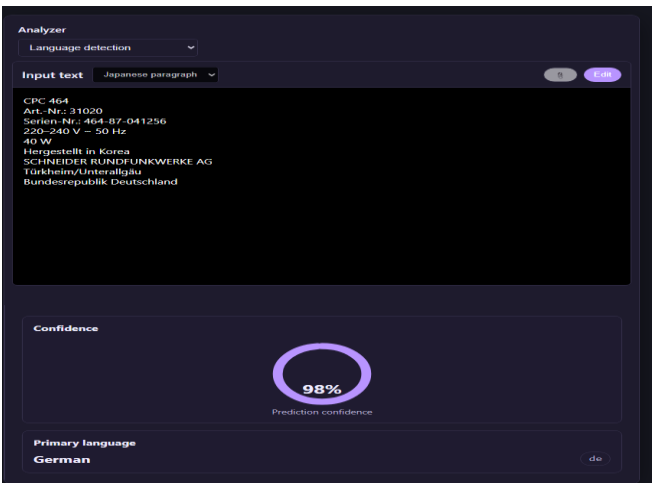


Fig. 3. German language detection output

As shown in Fig. 3, the Analyzer correctly identified the primary language as German (ISO code: de) with a 98% prediction confidence score. This result demonstrates that the language detection model relies on character-level n-grams and functional language patterns rather than getting misled by numbers, technical symbols, or model codes.

A third experiment tested a single input containing four distinct languages (English, French, German, and Spanish) within the same document to evaluate multi-language handling under complex conditions.

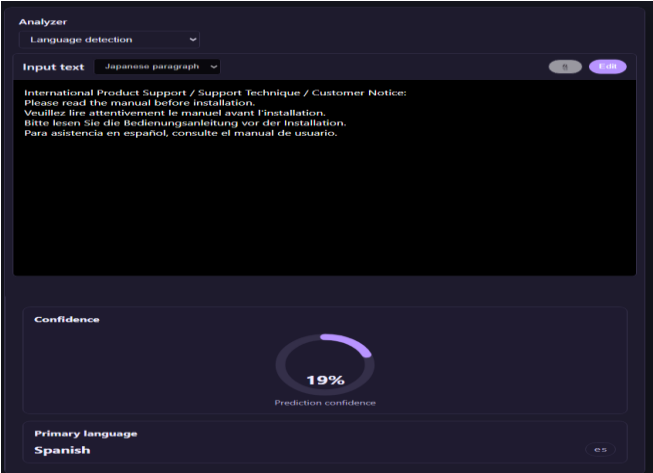


Fig. 4. Mixed multilingual language detection output

As shown in Fig. 4, the Analyzer selected Spanish (ISO code: es) as the primary language, but with a drastically reduced prediction confidence score of only 19%. This dramatic drop from 98% to 19% demonstrates a key limitation of standard document-level language detectors: when multiple languages are present, competing n-gram signals dilute the confidence of the model. This finding emphasizes the practical necessity of splitting multi-lingual documents into individual sentences or chunks before passing text into downstream NLP pipelines.

C. Text PII Extraction

The third task evaluated entity extraction capabilities using the Text PII Extraction Analyzer in the browser playground. The objective was to automatically detect sensitive Personally Identifiable Information (PII) within unstructured business documents to support privacy compliance workflows.

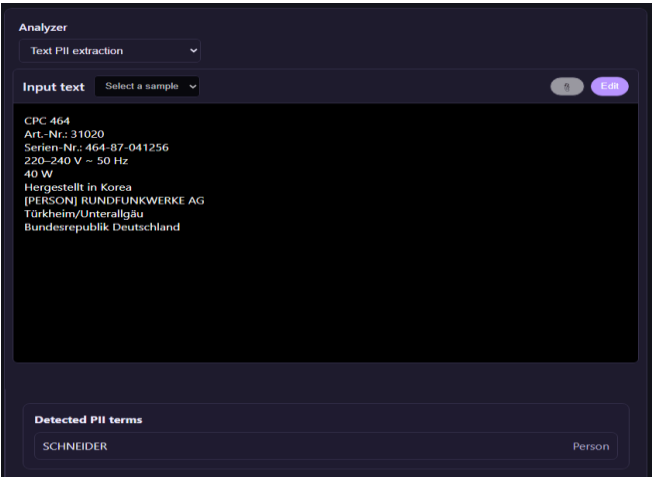


Fig. 5. Text PII extraction output

As shown in Fig. 5, the analyzer flagged "SCHNEIDER" as a Person entity and replaced it with [PERSON]. This result illustrates a common Named Entity Recognition (NER) false positive: because "Schneider" is a common German surname, the statistical model misidentified the brand name (SCHNEIDER RUNDfunkWERKE AG) as a personal name. This highlights that while automated PII extraction effectively sanitizes sensitive data, human oversight remains necessary to prevent corporate names from being erroneously redacted.

III. CORE LEARNING OUTCOMES

A. Tokenization Trade-Offs and Pre-Processing Impact

Tokenization converts raw text into numerical inputs for model processing. While word-level tokenization struggles with out-of-vocabulary terms and sparse matrices, modern subword algorithms balance vocabulary size and easily handle unseen words. Additionally, while classical statistical models rely on aggressive pre-processing like stop-word removal and stemming to reduce dimensionality, modern generative architectures require full sentence structures, punctuation, and casing to construct accurate attention maps.

B. TF-IDF vs. Dense Semantic Embeddings

Statistical methods like TF-IDF measure word importance using exact keyword frequency, making them computationally lightweight but unable to capture contextual meaning or synonyms. In contrast, dense semantic embeddings project tokens into a continuous vector space where metrics like cosine similarity measure conceptual proximity. This allows neural models to recognize underlying semantic relationships rather than relying strictly on character-level matches.

C. Context Vectors and Transformer Architectural Insights

Unlike early models that assigned static representations to words, modern Transformer architectures calculate dynamic context vectors that adjust a word's numerical representation based on surrounding text. These contextual vectors allow Small Language Models (SLMs) to resolve word ambiguity and evaluate long-range text dependencies. Executing these models locally proves that advanced tasks like summarization and entity extraction can run efficiently in privacy-preserving, client-side environments.

IV. COURSE APPLICATION & IMPACT

The core insights gained from this exercise establish a direct technical foundation for upcoming laboratory milestones and our capstone project architecture. Mastering foundational pre-processing, subword tokenization, and the

trade-offs between sparse TF-IDF matrices and dense semantic embeddings equips us with the criteria needed for selecting retrieval mechanisms in upcoming model optimization labs. Furthermore, practical experience with client-side Small Language Models, multilingual classification, and PII extraction demonstrates how advanced text analytics can be applied to real-world software. Recognizing real-world model limitations, such as false positives on brand names and confidence drops in mixed-language inputs, highlights the operational necessity of integrating privacy-by-design standards, sentence-level pre-processing, and human oversight into production-grade data pipelines.

V. CONCLUSION & FUTURE WORK

This laboratory report evaluated core NLP workflows within client-side AI playgrounds, demonstrating that local Small Language Models (SLMs) can effectively execute abstractive summarization, multilingual classification, and PII entity extraction without relying on external cloud infrastructure. Key experiential findings highlighted the operational necessity of sentence-level text chunking to prevent confidence drops in multi-language inputs, as well as human oversight to rectify statistical entity recognition false positives on brand names. Building upon these foundational takeaways, future study will focus on extending these concepts into modern Large Language Model (LLM) architectures by exploring Retrieval-Augmented Generation (RAG) pipelines for grounded document synthesis, parameter-efficient fine-tuning of lightweight SLMs, and hybrid rule-based entity extraction models to build scalable, privacy-preserving text analytics systems.

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